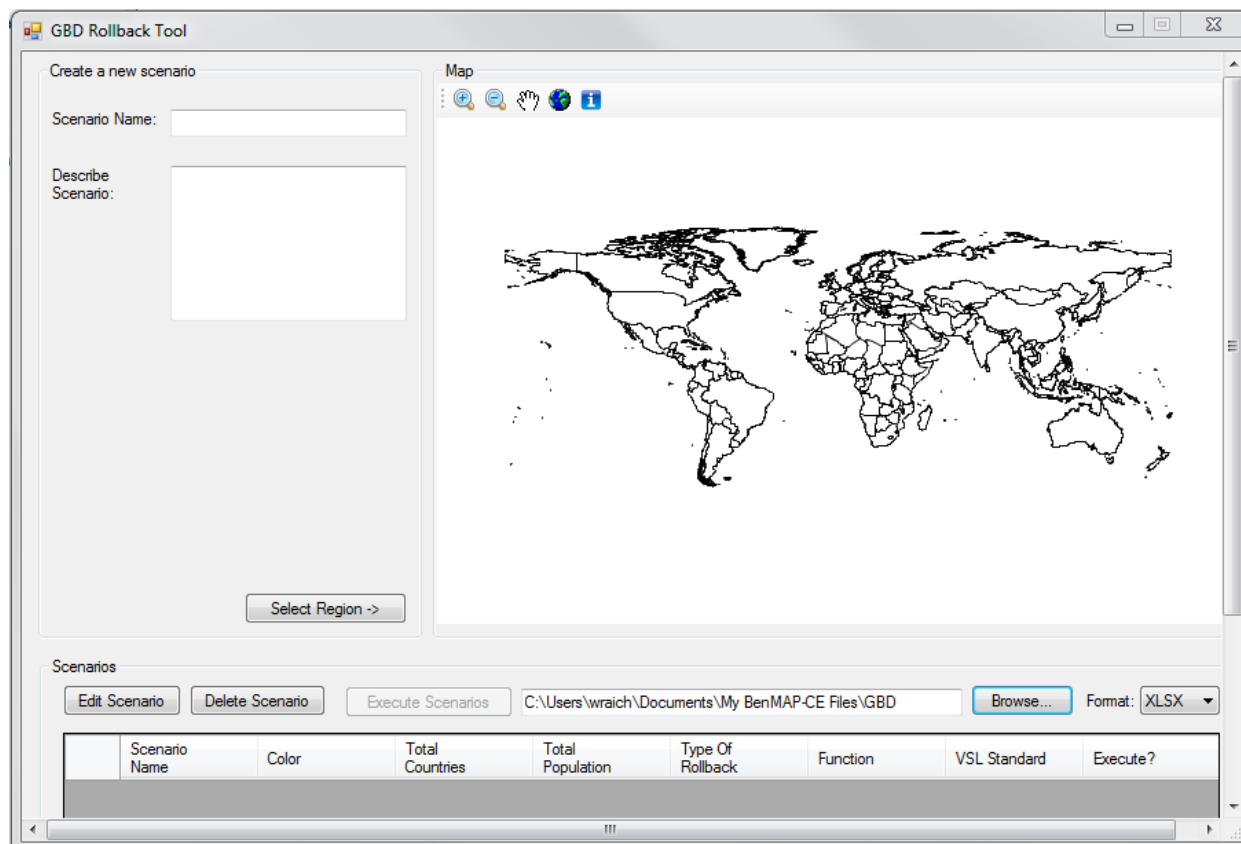


To begin, select *GBD Rollback* from the **Tools** menu. The **GBD Rollback Tool** window will be displayed.



9.7.1 Create New Scenario

To create a new scenario, first provide a **Scenario Name** (required) and description for the scenario (optional). Then, click the **Select Region** button.

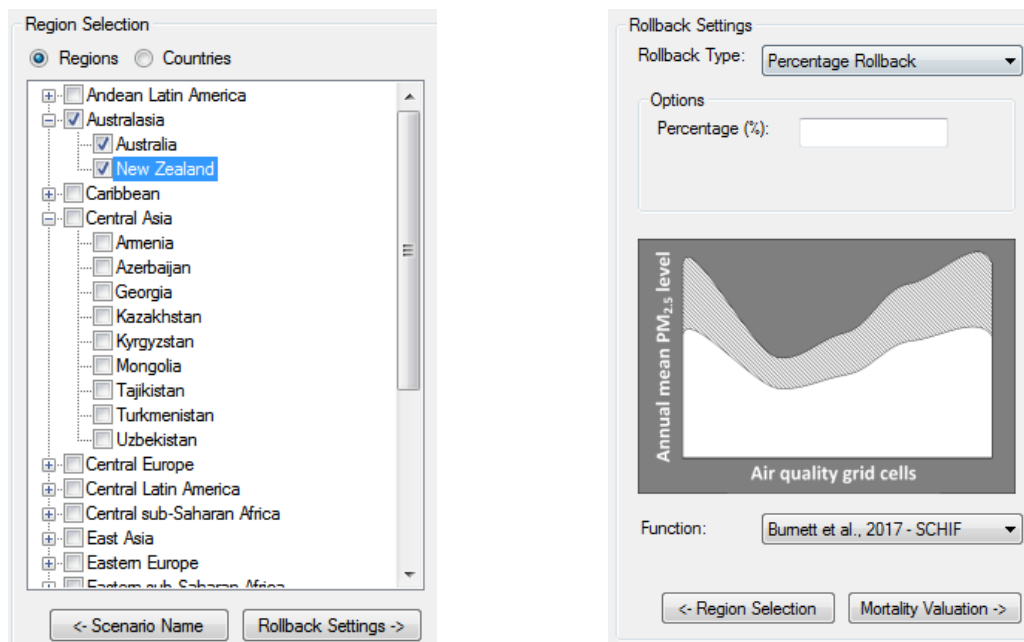
9.7.2 Select Regions or Countries

The **Region Selection** box provides options to view *Regions* (as defined in the GBD study) with the associated countries listed beneath, or to view *Countries*. Multiple selections are possible with either selection method and you can toggle between the two.

If you view *Regions*, you can select an entire region (all countries within this region) by checking the box next to the desired region. To view the individual countries for a region, click on the '+' sign to the left of the region. This will expand the tree menu. To collapse the tree menu, simply click on the '-' sign.

If you do not know the name of the region which contains the country you want to select, click the option to view *Countries*. You can search the alphabetically-sorted list to find countries.

As you make selections, the map will highlight them in a bright blue color. Note that the map is not interactive (i.e., you cannot click on the map to make selections). The toolbar located above the map allows you to zoom in and out, pan, view full extent, and identify countries.



9.7.3 Choose Rollback Settings

After selecting the countries or regions to analyze, click the **Rollback Settings** button.

The **Rollback Settings** pane allows you to choose the **Rollback Type** (*Percentage Rollback*, *Incremental Rollback*, or *Rollback to a Standard*) and configuration options.³³ (An illustrative example for the selected **Rollback Type** will be displayed below the *Options* box.) The tool allows for a negative rollback, indicating an increase in pollution concentration.

For *Percentage* and *Incremental Rollback* types, you must enter a percent value or whole number by which to reduce the pollutant concentration in all grid cells in the

Fundamental Concept - Rollback

A **rollback** is a simplified type of BenMAP analysis in which all monitor data are reduced under the Control scenario using a uniform rule. BenMAP-CE includes three rollback options for monitor data: **Percentage rollback** reduces all monitor observations by the same percentage. **Incremental rollback** reduces all observations by the same increment. **Rollback to a standard** reduces all monitor observations as necessary so that they all meet a specified standard.

³³ The tool uses a background PM_{2.5} concentration = 5.8 µg/m³ (lowest measured level in epidemiological literature).

selected countries. If you select *Rollback to a Standard*, you may select from a list of national pollutant standards enforced by various countries in different years³⁴.

Standard Group	Concentration Limit (µg/m ³)	Exposure Duration
WHO	10	Annual
US Primary	12	Annual
US Secondary	15	Annual
EU	25	Annual
Japan	15	Annual
China Class I	15	Annual
China Class II	35	Annual
Hong Kong	35	Annual
India	40	Annual
Singapore	12	Annual
WHO	25	24-hour
US Primary	35	24-hour
Japan	35	24-hour
China Class I	35	24-hour
China Class II	75	24-hour
Hong Kong	75	24-hour
India	60	24-hour
Singapore	37.5	24-hour

The **Function** dropdown allows users to select either the SCHIF (Burnett et al., 2017) or the IER function used in the 2013 GBD studies (Burnett et al., 2014).

9.7.4 Choose Mortality Valuation Settings

Once you have completed the Rollback Settings window, click the **Mortality Valuation** button. The **Mortality Valuation Settings** pane allows you to specify how you wish to calculate the economic benefits associated with avoided premature deaths.

Estimated reductions in premature mortality are valued using country-specific estimates of the value per statistical life (VSL). This value is not the monetary value of individual lives. Rather, it reflects the amount individuals are willing to pay to incrementally reduce their risks of death from adverse health conditions that may be caused by environmental pollution. For example, if each individual in a population

Fundamental Concept - VSL

The **Value of Statistical Life (VSL)** is the monetary value derived from economic studies of the amount of money individuals are willing to exchange for small reductions in their annual mortality risk, integrated over a whole population. It does NOT reflect the value of the life of any specific individual.

³⁴ Air Quality Standards / Guidelines / Objectives For Different Countries (updated April 2014). Table developed by Scott Voorhees, US EPA.

is willing to pay \$20 to reduce his or her risk of death by 10 in 100,000, the VSL for that population would be \$200,000 ($= \$20 \div (10/100,000)$). For 500 avoided deaths in that population, the economic benefit would be \$100,000,000 ($= 500 * \$200,000$).

The **VSL Standard** dropdown menu allows users to select the VSL estimate applied to mortality risk reductions. Because VSL estimates from primary research are not available in most countries, it is necessary to transfer international estimates across countries, adjusting for differences in income levels.³⁵ Users may select one of two base VSL estimates generously provided by the authors of World Bank (2016).³⁶ VSL estimates were derived by the authors using a benefit-transfer approach, with a base VSL of \$3.83 million (2011 USD) as recommended by a meta-analysis of willingness-to-pay (WTP) studies by the Organisation for Economic Co-operation and Development (OECD). The base VSL is transferred to other countries and years using the following formula:

$$VSL_{c,n} = VSL_{OECD} * \left(\frac{Y_{c,n}}{Y_{OECD}} \right)^e$$

where $VSL_{c,n}$ is the VSL for country c in year n ; VSL_{OECD} is the base OECD VSL; $Y_{c,n}$ is the GDP per capita for country c in year n ; Y_{OECD} is the average GDP per capita for the sample of OECD countries (roughly \$37,000); and e is the income elasticity of the VSL. In all instances, monetary values are adjusted for price inflation and converted to 2011 U.S. dollars at purchasing power parity (PPP) rates. While the authors provide VSL estimates for a range of income elasticity (e) values, we present their central results, which assume an elasticity of 1.2 for low- and middle-income countries, and 0.8 for high-income countries. This benefits transfer approach is also used with a base VSL estimate from USEPA of \$8.7 million (2011 USD).

Additionally, we use country-level VSL estimates to derive population-weighted average VSLs by GBD regions. These VSLs allow GBD Tool users to value reduced mortality at the regional level. The regional VSLs are also used to impute missing country-level VSLs for 29 of the 210 countries in the GBD tool. The “Metadata” tab in the GBD Tool Output spreadsheet displays VSL estimates by country and region.

The left pane of the **Mortality Valuation Settings** window provides a list of countries with their respective VSL estimates. Once you have selected the base VSL estimate (OECD or USEPA), click the **Save Scenario** button to save your rollback configuration. The map will update the color of the selected countries so that each saved scenario is uniquely different.

³⁵ Mortality risk reductions have been shown to be sensitive to income; individuals are willing to pay more for risk reductions as their income increases. Thus, the VSL is commonly adjusted to reflect differences in the average income levels both across countries and within a country over time.

³⁶ World Bank and IHME (2016). *The Cost of Air Pollution: Strengthening the Economic Case for Action*. Washington, DC: 2016 International Bank for Reconstruction and Development/The World Bank.